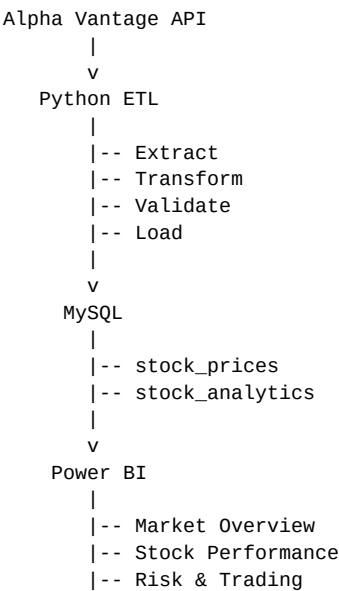


# Stock Market Analytics Project

## Complete Project Architecture & Data Flow

This document explains the complete architecture of the stock market analytics project, from data extraction through ETL, MySQL storage, analytics, Power BI dashboards, and scheduled automation.

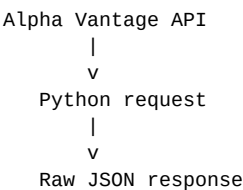
### 1. High-Level Architecture



### 2. Data Source — Alpha Vantage API

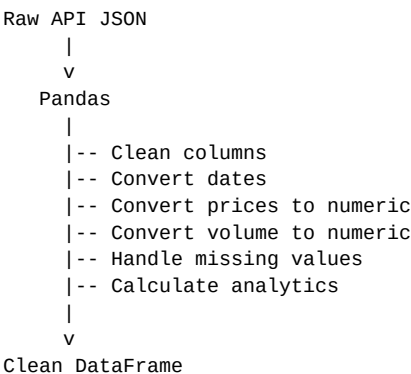
Alpha Vantage is the external data source for the project. Python requests stock-market data for the configured symbols. The current project uses RELIANCE.BSE, TCS.BSE, INFY.BSE, and IBM.

### 3. Extraction Layer



The extraction stage connects to the API, sends requests, receives the response, handles API-level errors where applicable, and passes the raw data to the transformation stage.

### 4. Transformation Layer

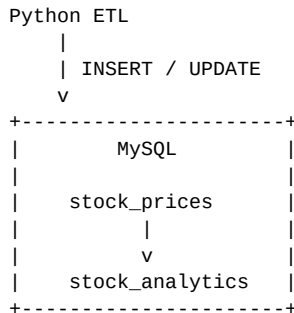


Python and Pandas convert the API response into structured tabular data suitable for database storage and analysis. This is a core part of the ETL process.

## 5. Data Validation

Before loading data into MySQL, the pipeline should validate the records. Typical checks include valid dates, numeric prices and volumes, valid stock symbols, missing values, and duplicate records.

## 6. MySQL Database Layer



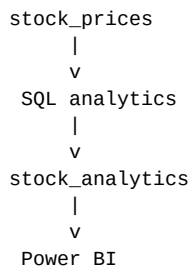
### stock\_prices

This is the main cleaned historical stock-price table. A typical structure contains fields such as id, symbol, date, open, high, low, close, and volume.

### stock\_analytics

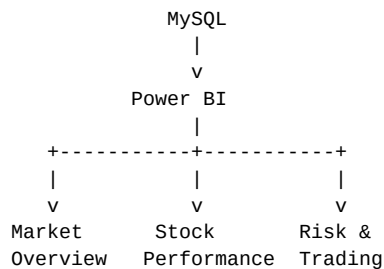
This analytical layer contains derived metrics that can be reused by reporting tools such as Power BI. Examples include returns, moving averages, volatility, price ranges, and other performance indicators.

## 7. Analytics Layer



The database can handle reusable analytical calculations, while Power BI focuses on interactive visualization, filtering, comparison, and presentation.

## 8. Power BI Layer



### Dashboard Page 1 — Market Overview

Provides a high-level view of the tracked stocks, including price information, market performance, volume, trends, and summary KPIs.

### Dashboard Page 2 — Stock Performance

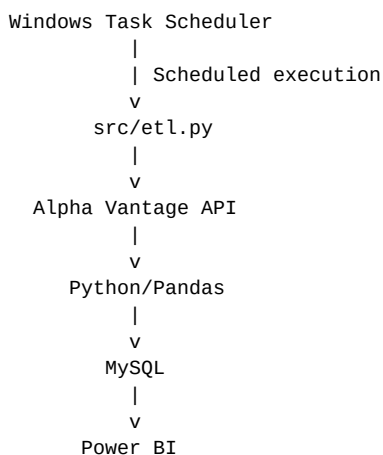
Focuses on individual stock behavior and comparison. It can show close-price trends, moving averages, returns, volume, and high/low values for RELIANCE, TCS, INFY, and IBM.

### Dashboard Page 3 — Risk & Trading

Focuses on volatility, returns, trading volume, and comparative risk/performance indicators.

## 9. Automation Layer

The remaining major step is scheduled ETL execution. Instead of manually running the ETL script, Windows Task Scheduler can launch the production ETL script on a defined schedule.



## 10. Recommended Project Folder Structure

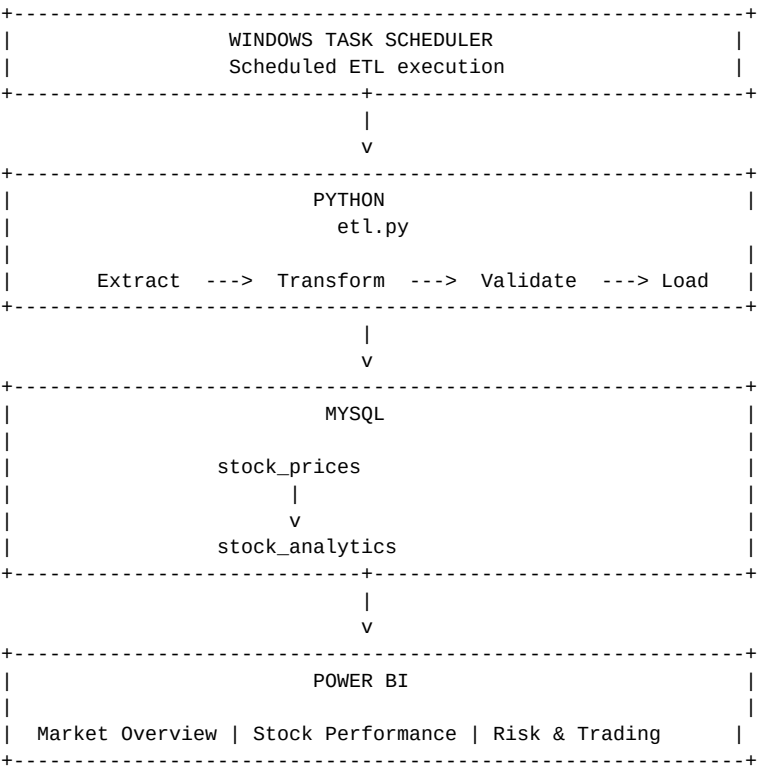
```
Stock-Market-Analytics/
|
+-- venv/
|
+-- src/
|   +-- extract.py
|   +-- test_stocks.py
|   +-- etl.py
|
+-- data/
|
+-- sql/
|
+-- dashboard/
|
+-- .env
|
```

```
+-- requirements.txt
|
+-- README.md
```

## 11. Technology Stack

| Layer           | Technology             | Purpose                     |
|-----------------|------------------------|-----------------------------|
| Data Source     | Alpha Vantage API      | Provides stock-market data  |
| Programming     | Python                 | ETL and pipeline logic      |
| Data Processing | Pandas                 | Cleaning and transformation |
| Database        | MySQL                  | Data storage and analytics  |
| Visualization   | Power BI               | Interactive dashboards      |
| Automation      | Windows Task Scheduler | Scheduled ETL execution     |

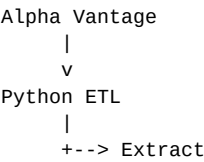
## 12. Complete Production Architecture



## 13. Interview Explanation

I built an end-to-end stock market analytics pipeline where Alpha Vantage provides stock data, Python performs extraction, transformation and validation, MySQL stores the cleaned data and analytical layer, and Power BI consumes the database to provide interactive market, performance, and risk dashboards. The ETL pipeline is designed to be automated through Windows Task Scheduler.

## 14. Final Data Flow



```
    +--> Transform
    +--> Validate
    +--> Load
    |
    v
MySQL
    |
    +--> stock_prices
    +--> stock_analytics
    |
    v
Power BI
    |
    +--> Market Overview
    +--> Stock Performance
    +--> Risk & Trading
    |
    v
Automated scheduled refresh via Windows Task Scheduler
```